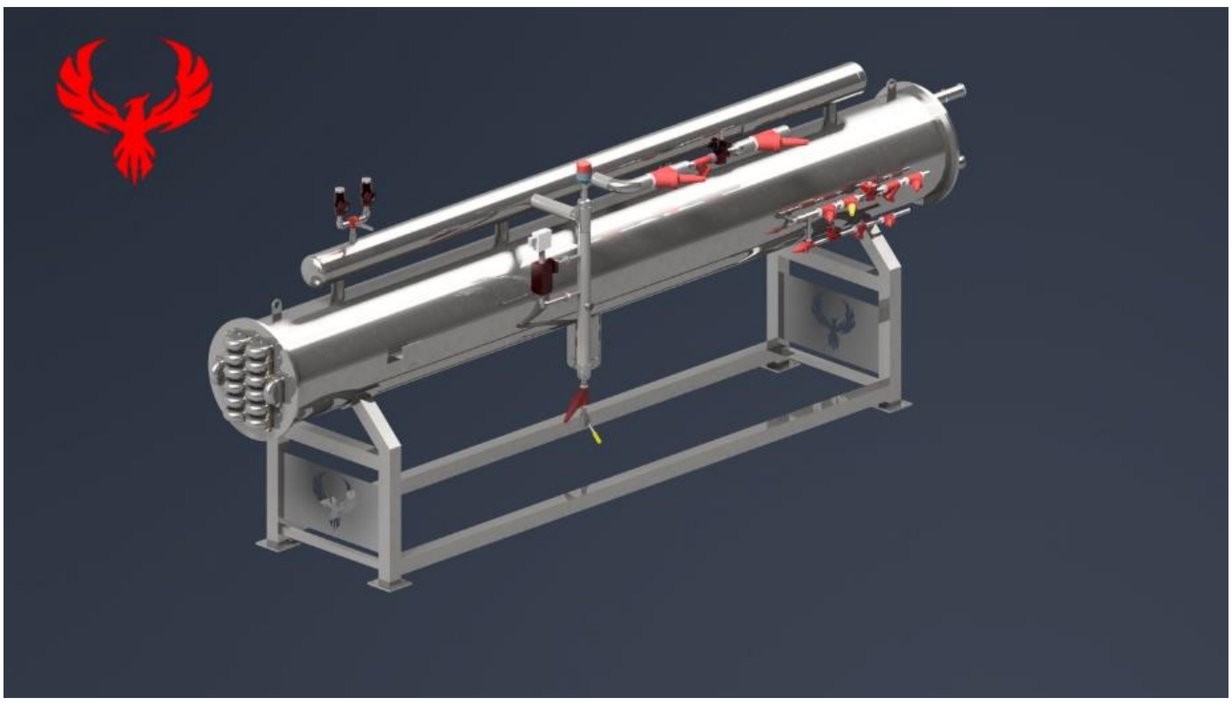
PHOENIX INNOVATIONS INC.

479-219-9100
Phoenix314.com

User Manual for Phoenix Innovations
Heat Exchanger
Sanderson Union Springs, AL



Serial # PPV-143
Model # PGI-010

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Contents

PHOENIX INNOVATIONS INC.

PHOENIX INNOVATIONS INC.

Letter of Conformity

Phoenix Innovations manufactures, sells, and services food processing equipment to exact customer specifications. Our equipment is designed and manufactured to meet or exceed both safety, hygiene, and environmental standards that will be present in the food processing industry. Equipment is manufactured using stainless steel, or other acceptable plastic components in food contact areas and uses corrosion resistant materials throughout the machine when possible. Equipment is designed and built with hygiene in mind and ease of both cleaning and maintaining.

RMA Procedures

A. When you receive your merchandise:
   a. Please open all boxes immediately and check contents. We must be notified of any damaged or defective products within 2 business days of you receiving your products. We also request that you also inspect all of your products for obvious defects/blemishes within 2 days of receiving them.

B. If you receive the incorrect product:
   a. It is very rare that you will receive an incorrect product; we double-check *all* orders before we ship them. However, if we made a mistake/error in shipping (it does rarely happen), please contact us. We will issue an RMA Number and return instructions on how and where to return the product. After receipt of the incorrect part we will then ship you the correct product and *pay the shipping costs both ways*. In some instances we may choose to ship a replacement overnight providing we receive another purchase order for the replacement. If you ordered the incorrect product and would like to return it, then please see the "Returns" section.

C. Shipping Damage:
   a. If the packaging appears damaged on the outside, <u>*please refuse to accept it from the carrier*</u>; or please make sure when signing the shipper's proof-of-delivery slip, you include a note stating the package is or appears damaged. If you do accept a damaged shipment, please contact us immediately. Depending on the product and where/how it was shipped, it may be necessary for you to deal with the carrier. Also, if there was any internal (non-visible) damage, this will need to be reported to us or the carrier within two days of receiving the product.

D. Returns:
   a. CRITERIA REQUIRED FOR RETURNING ITEM(S)
      i. Must be within 30 days from the day you received the item
      ii. Item must <u>not</u> have been installed
      iii. The item must be in perfect resalable condition in the original packaging
      iv. Custom orders or quoted items are non-returnable
      v. Item must have been shipped by a Common Carrier (UPS/FedEx/USPS) If the above criteria is met, then we will issue an RMA Number
   b. If you ordered the wrong products and wish to return the product, you will need to contact us to request an RMA Number. Returns will be subject to a 20% Restocking fee.
   c. Please do NOT return any product that you have ordered without contacting us for an RMA Number first. This will make processing your credit very difficult and may delay your credit.

E. Returned Materials Authorization:
   a. When it is necessary to return a product, please contact us to request an RMA Number. All returns, regardless of how they were shipped or where they were shipped from must be
   referenced by an RMA Number. Packages returned to us without an RMA number will not be returned to you. Handling returns in the manner that we do is necessary to ensure that you receive proper credit in a timely fashion.
   b. When you request an RMA Number to return a product, we will either issue the RMA at that time, or contact the warehouse that it shipped from and request that one be issued. If we need to receive the RMA from another warehouse, we will let you know by forwarding the information to you along with shipping instructions as soon as we receive it. The instructions for return must be followed carefully.
   c. Once the product has been received at our warehouse, it will be inspected and you will be credited for the returned product less the applicable Restocking Fee and original shipping and processing charges.
   d. If the product is returning to another warehouse, they will contact us once they have received the product, inspected it, and we have been issued credit. Once we have been credited for the return, your credit will be issued less the Restocking Fee and original shipping and processing charges.

Safety

The Control of Hazardous Energy (Lockout/Tagout)

The following lockout/tagout program is provided only as a guide to assist employers and employees in complying with the requirements of 29 CFR 1910.147, as well as to provide other helpful information. It is not intended to supersede the requirements of the standard. An employer should review the standard for particular requirements which are applicable to their individual situation and make adjustments to this program that are specific to their company. An employer will need to add information relevant to their particular facility in order to develop an effective, comprehensive program.

Lockout/Tagout Procedure

OBJECTIVE

The objective of this procedure is to establish a means of positive control to prevent the accidental starting or activating of machinery or systems while they are being repaired, cleaned and/or serviced. This program serves to:

A. Establish a safe and positive means of shutting down machinery, equipment and systems.
B. Prohibit unauthorized personnel or remote control systems from starting machinery or equipment while it is being serviced.
C. Provide a secondary control system (tagout) when it is impossible to positively lockout the machinery or equipment.
D. Establish responsibility for implementing and controlling lockout/tagout procedures.
E. Ensure that only approved locks, standardized tags and fastening devices provided by the company will be utilized in the lockout/tagout procedures.

ASSIGNMENT OF RESPONSIBILITY

A. ***Responsible Person*** will be responsible for implementing the lockout/tagout program.
B. ***Responsible Persons*** are responsible for enforcing the program and insuring compliance with the procedures in their departments.
C. ***Responsible Person*** is responsible for monitoring the compliance of this procedure and will conduct the annual inspection and certification of the authorized employees.
D. ***Authorized employees*** (those listed in Attachment A) are responsible for following established lockout/tagout procedures. An authorized employee is defined as a person who locks out or tags out machines or equipment in order to perform servicing or maintenance on that machine or equipment. An affected employee becomes an authorized employee when that employee's duties include performing servicing or maintenance covered under 1910.147, The Control of Hazardous Energy (lockout/tagout).
E. ***Affected employees*** (all other employees in the facility) are responsible for insuring they do not attempt to restart or re-energize machines or equipment that are locked out or tagged out. An affected employee is defined as a person whose job requires him/her to operate or use a machine or equipment on which servicing or maintenance is being performed under lockout or tagout, or whose job requires him/her to work in an area in which such servicing or maintenance is being performed.

PROCEDURES

The ensuing items are to be followed to ensure both compliance with the OSHA Control of Hazardous Energy Standard and the safety of our employees.

Employees who are required to utilize the lockout/tagout procedure (see Attachment A) must be knowledgeable of the different energy sources and the proper sequence of shutting off or disconnecting energy means. The four types of energy sources are:

1. electrical (most common form);
2. hydraulic or pneumatic;
3. mechanical (including gravity).

More than one energy source may be utilized on some equipment and the proper procedure must be followed in order to identify energy sources and lockout/tagout accordingly. See Attachment F for specific procedure format.

ELECTRICAL

1. Shut off power at machine and disconnect.
2. Disconnecting means must be locked or tagged.
3. All controls must be returned to their safest position.
4. Points to remember:
    a. If a machine or piece of equipment contains capacitors, they must be drained of stored energy.
    b. Possible disconnecting means include the power cord, power panels (look for primary and secondary voltage), breakers, the operator's station, motor circuit, relays, limit switches, and electrical interlocks.
    c. Some equipment may have a motor isolating shut-off and a control isolating shut-off.
    d. If the electrical energy is disconnected by simply unplugging the power cord, the cord must be kept under the control of the authorized employee or the plug end of the cord must be locked out or tagged out.

HYDRAULIC / PNUEMATIC

1. Shut off all energy sources (pumps and compressors). If the pumps and compressors supply energy to more than one piece of equipment, lockout or tagout the valve supplying energy to the piece of equipment being serviced.
2. Stored pressure from hydraulic/pneumatic lines shall be drained/bled when release of stored energy could cause injury to employees.
3. Make sure controls are returned to their safest position (off, stop, standby, inch, jog, etc.).

MECHANICAL ENERGY

Mechanical energy includes gravity activation, energy stored in springs, etc.

1. Block out or use die ram safety chain.
2. Lockout or tagout safety device.
3. Shut off, lockout or tagout electrical system.
4. Check for zero energy state.
5. Return controls to safest position.

Release from Lockout/Tagout

1. Inspection: Make certain the work is completed and inventory the tools and equipment that were used.
2. Clean-up: Remove all towels, rags, work-aids, etc.
3. Replace guards: Replace all guards possible. Sometimes a particular guard may have to be left off until the start sequence is over due to possible adjustments. However, all other guards should be put back into place.
4. Check controls: All controls should be in their safest position.
5. Check the Work Area: The work area shall be checked to ensure that all employees have been safely positioned or removed and notified that the lockout/tagout devices are being removed.
6. Remove locks/tags: Remove only your lock and / or tag.

Service or Maintenance Involving More than One Person

When servicing and/or maintenance is performed by more than one person, each authorized employee shall place his own lock or tag on the energy isolating source. This shall be done by utilizing a multiple lock scissors clamp if the equipment is capable of being locked out. If the equipment cannot be locked out, then each authorized employee must place his tag on the equipment.

Removal of an Authorized Employee's Lockout/Tagout by the Company

Each location must develop written emergency procedures that comply with 1910.147(e)(3) to be utilized at that location. Emergency procedures for removing lockout/tagout should include the following:

1. Verification by employer that the authorized employee who applied the device is not in the facility.
2. Make reasonable efforts to advise the employee that his/her device has been removed. (This can be done when he/she returns to the facility).
3. Ensure that the authorized employee has this knowledge before he/she resumes work at the facility.
4. **Shift or Personnel Changes**

Shift or Personnel Changes

Each facility must develop written procedures based on specific needs and capabilities. Each procedure must specify how the continuity of lockout or tagout protection will be ensured at all times. See 1910.147(f)(4).

Procedures for Outside Personnel/Contractors

Outside personnel/contractors shall be advised that the company has and enforces the use of lockout/tagout procedures. They will be informed of the use of locks and tags and notified about the prohibition of attempts to restart or re-energize machines or equipment that are locked out or tagged out.

The company will obtain information from the outside personnel/contractor about their lockout/tagout procedures and advise affected employees of this information.

The outside personnel/contractor will be required to sign a certification form (see Attachment E). If outside personnel/contractor has previously signed a certification that is on file, additional signed certification is not necessary.

Training and Communication

Each authorized employee who will be utilizing the lockout/tagout procedure will be trained in the recognition of applicable hazardous energy sources, type and magnitude of energy available in the work place, and the methods and means necessary for energy isolation and control.

Each affected employee (all employees other than authorized employees utilizing the lockout/tagout procedure) shall be instructed in the purpose and use of the lockout/tagout procedure, and the prohibition of attempts to restart or re-energize machines or equipment that are locked out or tagged out.

Training will be certified using Attachment B (Authorized Personnel) or Attachment C (Affected Personnel). The certifications will be retained in the employee personnel files.

Periodic Inspection

A periodic inspection (at least annually) will be conducted of each authorized employee under the lockout/tagout procedure. This inspection shall be performed by the ***(Responsible person)*___**. If ***(Responsible person)*___** is also using the energy control procedure being inspected, then the inspection shall be performed by another party.

The inspection will include a review between the inspector and each authorized employee of that employee's responsibilities under the energy control (lockout/tagout) procedure. The inspection will also consist of a physical inspection of the authorized employee while performing work under the procedures.

The__***(Responsible person)*___** shall certify in writing that the inspection has been performed. The written certification (Attachment D) shall be retained in the individual's personnel file.

Attachments:

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ATTACHMENT A

List of Authorized Personnel
For Lockout/Tagout Procedures

Name	Job Title

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ATTACHMENT B

Certification of Training
(Authorized Personnel)

I certify that I received training as an authorized employer under ***Company Name*** Lockout/Tagout program. I further certify that I understand the procedures and will abide by those procedures.

AUTHORIZED EMPLOYEE SIGNATURE DATE

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ATTACHMENT C

Certification of Training
(Affected Personnel)

I certify that I received training as an Affected Employee under ***Company Name*** Lockout/Tagout Program. I further certify and understand that I am prohibited from attempting to restart or re-energize machines or equipment that are locked out or tagged out.

AUTHORIZED EMPLOYEE SIGNATURE DATE

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ATTACHMENT D

Lockout/Tagout Inspection Certification

I certify that ____***Equipment***____ was inspected on this date utilizing lockout/tagout procedures. The inspection was performed while working on ____***Equipment***____ .

AUTHORIZED EMPLOYEE SIGNATURE DATE

INSPECTOR SIGNATURE DATE

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ATTACHMENT E

Outside Personnel/Contractor Certification

I certify that ____________________________ and ______________________ (outside personnel/contractor) have informed each other of our respective lockout/tagout procedures.

AUTHORIZED EMPLOYEE SIGNATURE DATE

INSPECTOR SIGNATURE DATE

PHOENIX INNOVATIONS INC.

ATTACHMENT F

Equipment Specific Procedure
for

____

Date: ________________________________

Machine Identification

General Description: ____
____
____
____

Manufacturer: ____
____
____
____

Model Number:____
____

Serial Number: ____
____

** If more than one piece of same equipment, list all serial numbers.*

Location of equipment: ____
____
____
____

Operator Controls

The types of controls available to the operator need to be determined. This should help identify energy sources and lockout capacity for the equipment.

List types of operator controls: ____
____
____
____

Energy Sources

The energy sources, such as electrical, steam, hydraulic, pneumatic, natural gas, stored energy, etc.) present on this equipment are:

ENERGY SOURCE	LOCATION	Lockable		Type lock or block needed
		Yes	No	

Machine Operation

Start Up

It is always important to follow start-up procedures in the proper order to ensure that both personnel and equipment are not damaged. Always follow safety guidelines when operating electrical equipment and refrigeration equipment.

The units are shipped without valves and must be supplied and piped by others. The liquid and suction refrigerants are needed to operate correctly. Check to be sure damage did not occur during shipping.

Close liquid and suction hand valves. The suction regulator and liquid solenoids should be opened manually to check for leaks. Refrigerant pressure of 60-70 lbs. is adequate to verify if any leaks exist. Place valves back in automatic position after testing.

NOTE 1: The dual relief valve is set at 250 psig and this pressure should not be exceeded. The dual valves must be piped outside in an approved manner that is in accordance with local codes (Ref ASHRAE 15).

NOTE 2: Make sure suction line hand valves are open before any product is pumped through the heat exchanger. Failure to do so can result in excessive pressure build up and cause the relief valves to blow.

Before turning the unit on, verify the temperature and pressure readings are as expected. The pump supplying product should also be on. The Target Temp button is pressed to change the desired outlet temperature. Turn the run switch to run.

During Operation

Once the unit is switched on, if the target temp is lower than the current temp, and enough inlet pressure is sensed, the Liquid valve will open, and the suction regulator will kick to the low side. Once the desired temperature is reached, the Liquid and Suction vales will actuate accordingly to maintain that temperature.

The minimum inlet pressure keeps the unit from “running dry” while there is no product flowing through the tubes. Due to pump placement, each unit will have a different pressure under normal operation, and this will need to be adjusted accordingly.

The maximum inlet pressure determines when the product is “freezing up”. Adjusting this will keep the unit from cooling when this pressure is detected at the inlet.

The “Hot Gas Gap” is an adjustable offset that determines at what temperature the hot gas is triggered. For example, if the target temperature is set at 34°, and the hot gas gap is set to 2.0, if the product temperature dropped below 32°, the hot gas would cycle to avoid a freeze up.

Shut Down

To shut down the Heat exchanger simply turn the run switch to off.

Freeze Protection

Pressure Based Protection

The Phoenix Innovations series of shell and tube heat exchangers are designed to prevent freeze-up conditions from occurring. This is accomplished with continuous monitoring of the product inlet pressure and both inlet and outlet product temperatures. During the commissioning phase of the installation of a Phoenix Innovations heat exchanger, your field technician will establish a base line product inlet pressure, with the absence of refrigeration applied. The afore-mentioned inlet pressure will now be utilized as our “baseline product inlet pressure”, at this point the field technician will establish both a high and low product inlet pressure. Once these set points are established, this will complete the product inlet pressure set up.

Theory of Operation

During normal operation of the Phoenix Innovations shell and tube heat exchanger, the PLC will continuously monitor product inlet pressure. At the detection of an elevated product inlet pressure, the PLC will deactivate the liquid refrigerant feed solenoid, switch the suction regulator to its highest set-point (above 55PSIG), and "burst' activate the hot gas solenoid until the product inlet pressure returns to normal operating parameters. Your operator will be notified that the system is in defrost by a notification banner on the HMI screen.

Temperature Based Protection

The Phoenix Innovations series of shell and tube heat exchangers are designed for simple, accurate temperature control. This is accomplished with continuous monitoring of both inlet and outlet product temperatures. During the commissioning phase of the installation of a Phoenix Innovations heat exchanger, your field technician will establish a calibrated inlet and outlet temperatures readings within the HMI. These temperatures readings will now be utilized as our "baseline product inlet and outlet temperature readings", at this point the field technician will establish both target temperature and a high limit for product temperatures. Once these set points are established, this will complete the product temperature set up.

Theory of Operation

During normal operation of the Phoenix Innovations shell and tube heat exchanger, the PLC will continuously monitor product inlet/outlet temperatures. At the detection that the product has achieved the outlet temperature set point, the PLC will deactivate the refrigeration suction valve. This will in turn switch the suction regulator to its highest set-point (above 55PSIG), allowing the outlet product temperature to rise above the temperature set-point preventing freeze-up conditions. Once the product's outlet temperature achieves the high limit temperature setting then the PLC re-activates the refrigeration suction regulator allowing the product to achieve outlet temperature set-point. In the event the product temperature set-point is achieved, and the outlet temperature continues to decline, the PLC will activate the described "Freeze-up protection", described in Pressure Based Protection.

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Preventative Maintenance

Monthly

	Date	Initial
1. Inspect control cabinet gaskets for wear/damage	___________/	___________
2. Check cabinets for water ingress	___________/	___________
3. Inspect pump contactors for scorching/arcing. Replace if necessary ..	___________/	___________
4. Inspect all covers and gaskets on the machine.	___________/	___________
5. Check for oil and drain if necessary	___________/	___________
6. Check/Calibrate RTDs..	___________/	___________

Annually

7. Check Vari Level to Verify accuracy	___________/	___________
a. Manually fill vessel until Ammonia is in the center of sight glass...	___________/	___________
b. Ammonia level should read 50%..	___________/	___________
c. If calibration is off use the System Setup Page to set 50%	___________/	___________

Signed .... ______________________

Date... ______________________

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Controls

PLC (Programmable Logic Controller)

Micro 800 PLC

Programming

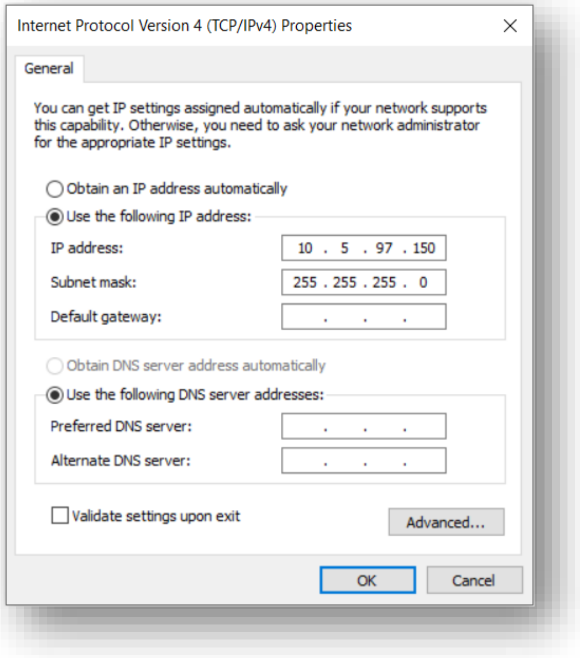
Programming a new Micro800 with Backup Card

Our programs are set to "Load On Startup", so every time power is applied to the PLC, the program will be pulled from the memory card and saved to the PLC. After installing the new Micro800, the memory card adapter should be in the first expansion port on the front of the PLC. Once power is applied, the program should bepulled from the memory card.

Programming With Connected Componnents Workbench

The Micro800 series of PLC's are programmed with the program Connected Components Workbench (CCW). This can be downloaded for free on Rockwell Automation's website. These instructions show how to connect via an ethernet cord.

1. Configure the ethernet adapter settings on the computer you will be using to communicate with the Micro800.
    a. On Windows 10, right click the windows icon and select "Network Connections", then click "Ethernet".
    b. On the right side of the window, select "Change Adapter Options".
    c. Right click on the ethernet adapter you will be using, then click "Properties".
    d. Select "Internet Protocol Version 4" and click "Properties".
    e. Change the settings to match the picture below.



(These changes can be undone to restore normal functionality of your ethernet port)

2. Connect an ethernet cord between the PLC and computer, and apply power tothe PLC.
3. If you intend to alter the program, switch the Run/Program switch to programon the bottom left corner of the PLC.
4. You can use RSLinx (part of the CCW install) to browse the PLC’sin the range you configured in step 1.e. to confirm communication between the PLC and computer.

   (The default IP of our PLC is 10.5.97.5, and the HMI is 10.5.97.10)

If you have experience with Studio 5000, follow this link for help with Connected Components Workbench
https://literature.rockwellautomation.com/idc/groups/literature/documents/qr/9328-qr001_-en-e.pdf

Device Configuration

Controller

Overview

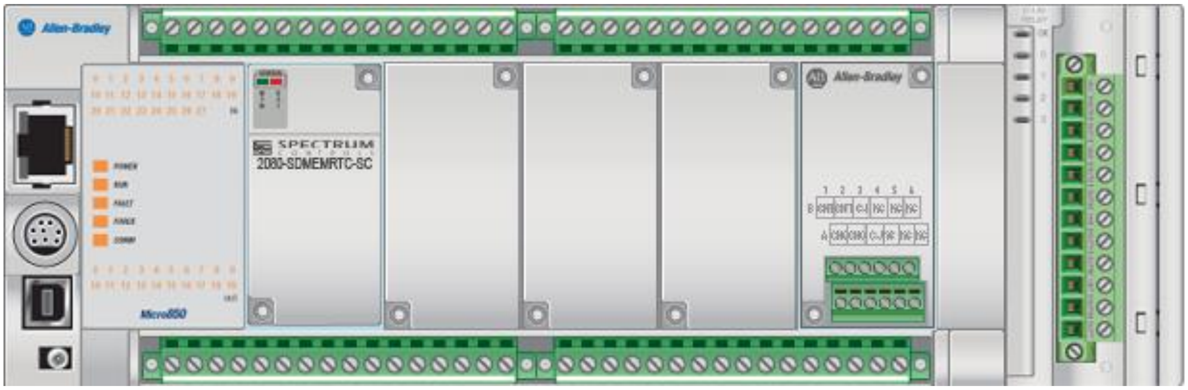


Figure 1 Your Micro850 with expansion modules

General

Name	Description	Vendor Name	Catalog ID	Product Lifecycle	Controller Project Version	Download Source Code
Micro850		Allen-Bradley	2080-LC50-48QBB	Discontinued	12	Yes

Plug-in Modules

Physical Slot	Catalog ID	When module is not present
1	2080-SDMEMRTC-SC	Do not fault controller (optional module)
5	2080-RTD2	Fault controller (required module)

Expansion Modules

Physical Slot	Catalog ID	When module is not present
1	2085-IF4	Fault controller (required module)

Startup/Faults

Mode Behavior	Fault Override	Memory Module	Hard Fault
Retain previous power-down mode	Do not clear fault	2080-SDMEMRTC-SC	Stop controller

Serial Port

Common Settings

Driver	Baud Rate	Parity	Station Address
CIP Serial	38400	None	1

Protocol Control

DF1 Mode	Media	Control Line	Error Detection
Full Duplex	RS232	No Handshake	CRC

Embedded Responses	Duplicate Packet Detection	ACK Timeout	NAK Retries
After One Received	Enabled	50	3

ENQ Retries

3	

USB Port

Status	Suspend State	Bus Speed	Mode	State
Enabled	Not Suspended	Full Speed	Slave Mode	Ready

Ethernet

Port Settings
Port State: Enabled
Auto-Negotiate Speed and Duplex Mode: Enabled

Internet Protocol (IP) Settings

Startup Configuration	IP Address	Subnet Mask	Gateway Address	Detect duplicate IP address
Static	10.5.97.7	255.255.255.0	10.5.97.100	Enabled

EtherNet/IP
Inactivity Timeout: 120 sec
Port State: Enabled
Auto-Negotiate Speed and Duplex Mode: Enabled

Plug-in Modules

2080-SDMEMRTC-SC

General

Catalog ID	Vendor Name	Description	Product Type	Slot	Revision
2080-SDMEMRTC-SC	Spectrum Controls	microSD card and high accuracy RTC	Specialty plug-in	1	1

Configuration

Memory Card Settings

Load on power up	Overwrite Ethernet Settings	Include Project & Logical values upon Backup/Restore
Load Always	True	True

2080-RTD2

General

Catalog ID	Vendor Name	Description	Product Type	Slot	Revision
2080-RTD2	Allen-Bradley	2-channel, non-isolated RTD module	Analog plug-in	5	1

Input

Label	Variable Name
Input 0	IO_P5_AI_00
Input 1	IO_P5_AI_01
Input 2	IO_P5_AI_02
Input 3	IO_P5_AI_03
Input 4	IO_P5_AI_04

Configuration

Channel	RTD Type	Data Update Rate
0	100 Pt 385	16.7 Hz
1	100 Pt 385	16.7 Hz

Expansion Modules

2085-IF4

General

Catalog ID	Vendor Name	Description	Product Type	Slot	Revision
2085-IF4	Allen-Bradley	4-channel 14bit analog voltage and current input module	Multi-Channel Analog	1	1

Input

Label	Variable Name
Input 0	IO_X1_AI_00
Input 1	IO_X1_AI_01
Input 2	IO_X1_AI_02
Input 3	IO_X1_AI_03

Status

Label	Variable Name
Status 0	IO_X1_ST_00
Status 1	IO_X1_ST_01
Status 2	IO_X1_ST_02

Channel 0

Enable Channel: Enabled

Min-Max Input Range	Data Format	Input Filter
4mA to 20mA	Engineering Units	50/60Hz Rejection

Alarm Limits

High High Alarm	High Alarm	Low Alarm	Low Low Alarm
Disabled	Enabled	Enabled	Disabled
21 mA	20 mA	4 mA	3.2 mA

Channel 1

Enable Channel: Enabled

Min-Max Input Range	Data Format	Input Filter
4mA to 20mA	Engineering Units	50/60Hz Rejection

Alarm Limits

High High Alarm	High Alarm	Low Alarm	Low Low Alarm
Disabled	Enabled	Enabled	Disabled
21 mA	20 mA	4 mA	3.2 mA

Channel 2

Enable Channel: Enabled

Min-Max Input Range	Data Format	Input Filter
4mA to 20mA	Engineering Units	50/60Hz Rejection

Alarm Limits

High High Alarm	High Alarm	Low Alarm	Low Low Alarm
Disabled	Enabled	Enabled	Disabled
21 mA	20 mA	4 mA	3.2 mA

Channel 3

Enable Channel: Disabled

Min-Max Input Range	Data Format	Input Filter
4mA to 20mA	Engineering Units	50/60Hz Rejection

Alarm Limits

High High Alarm	High Alarm	Low Alarm	Low Low Alarm
Disabled	Enabled	Enabled	Disabled

21 mA	20 mA	4 mA	3.2 mA

HMI (Human Machine Interface)
HMI (Human Machine Interface)

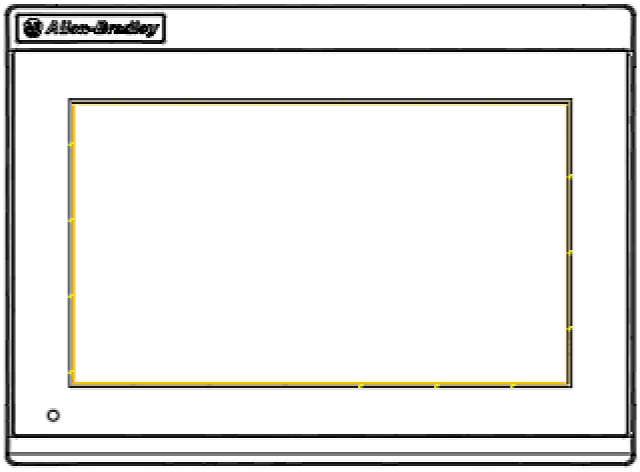


Figure 2 PanelView 800 7" HMI

Configuration

General

Name	Description	Vendor Name	Catalog ID	Controller Project Version	Download Source Code
PanelView 800		Allen-Bradley	271R-T7T	8	Yes

Ethernet

Port Settings
Port State: Enabled
Auto-Negotiate Speed and Duplex Mode: Enabled

Internet Protocol (IP) Settings

Startup Configuration	IP Address	Subnet Mask	Gateway Address	Detect duplicate IP address
Static	10.5.97.11	255.255.255.0	0.0.0.0	Enabled

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Heat Exchanger Home

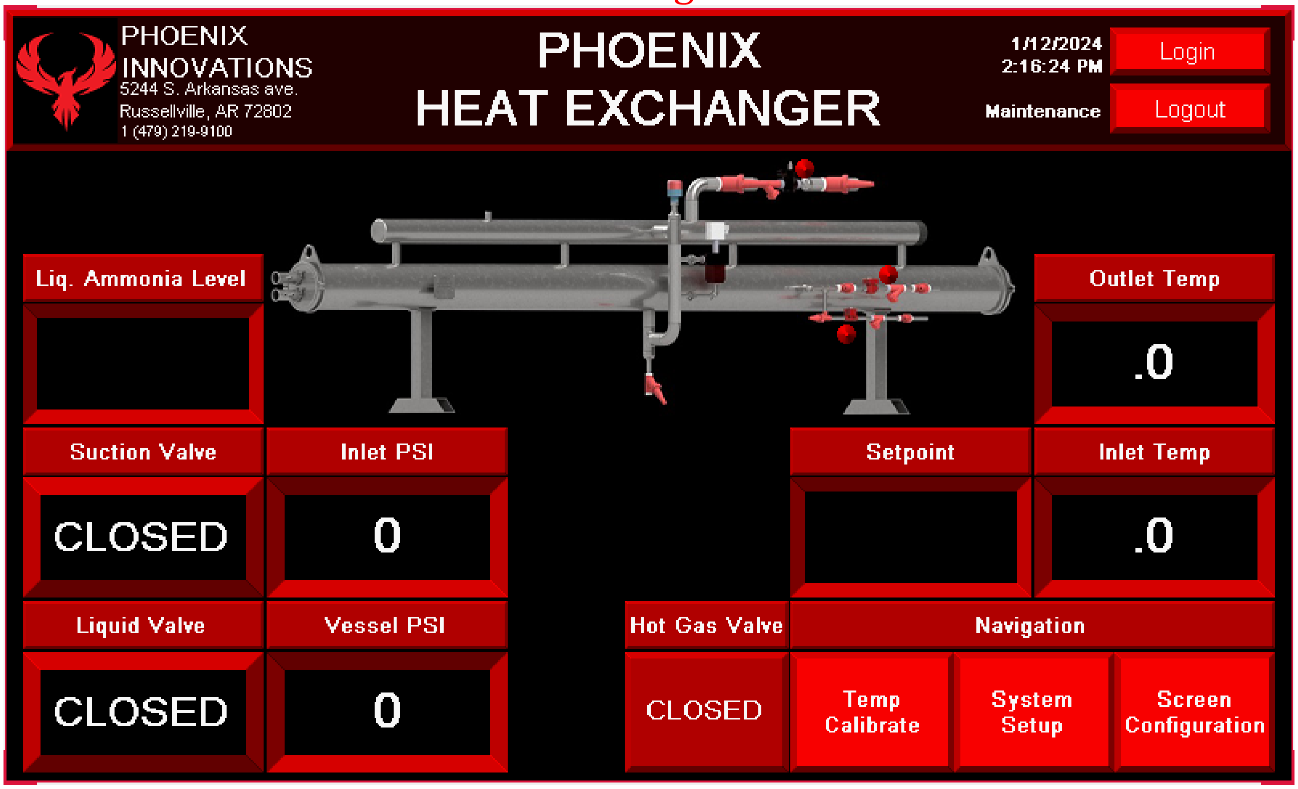

Figure 3 Home Screen Preview

- Liq. Ammonia Level – Displays the level of liquid ammonia in the vessel of the heat exchanger.
- Suction valve – Displays the status of the suction valve. *OPEN/CLOSED*
- Liquid Valve – Displays the status of the liquid valve. *OPEN/CLOSED*
- Pump Inlet PSI – Displays the pressure at the inlet of the Heat Exchanger.
- Vessel PSI – Displays the ammonia pressure of the vessel.
- Outlet Temp – Shows the water temperature at the outlet of the Heat Exchanger. This is the water temperature the HEX is supplying to the Process.
- Inlet Temp – Shows the water temperature at the inlet of the Heat Exchanger. This is the water temperature the HEX is receiving from the Process.
- Setpoint – Shows setpoint of the Heat Exchanger.
- Hot Gas Valve – Displays the status of the hot gas valve. *OPEN/CLOSED*
- Navigation – This panel is for navigating to the different screens.
  - Temp Calibrate – LOGIN REQ. – Navigates to the temperature / pressure calibration screen
  - System Setup – LOGIN REQ. – Navigates to the first configuration page.
  - Screen Configuration – LOGIN REQ. – Navigates to the built in terminal settings workspace.

PHOENIX INNOVATIONS INC.

System Configuration

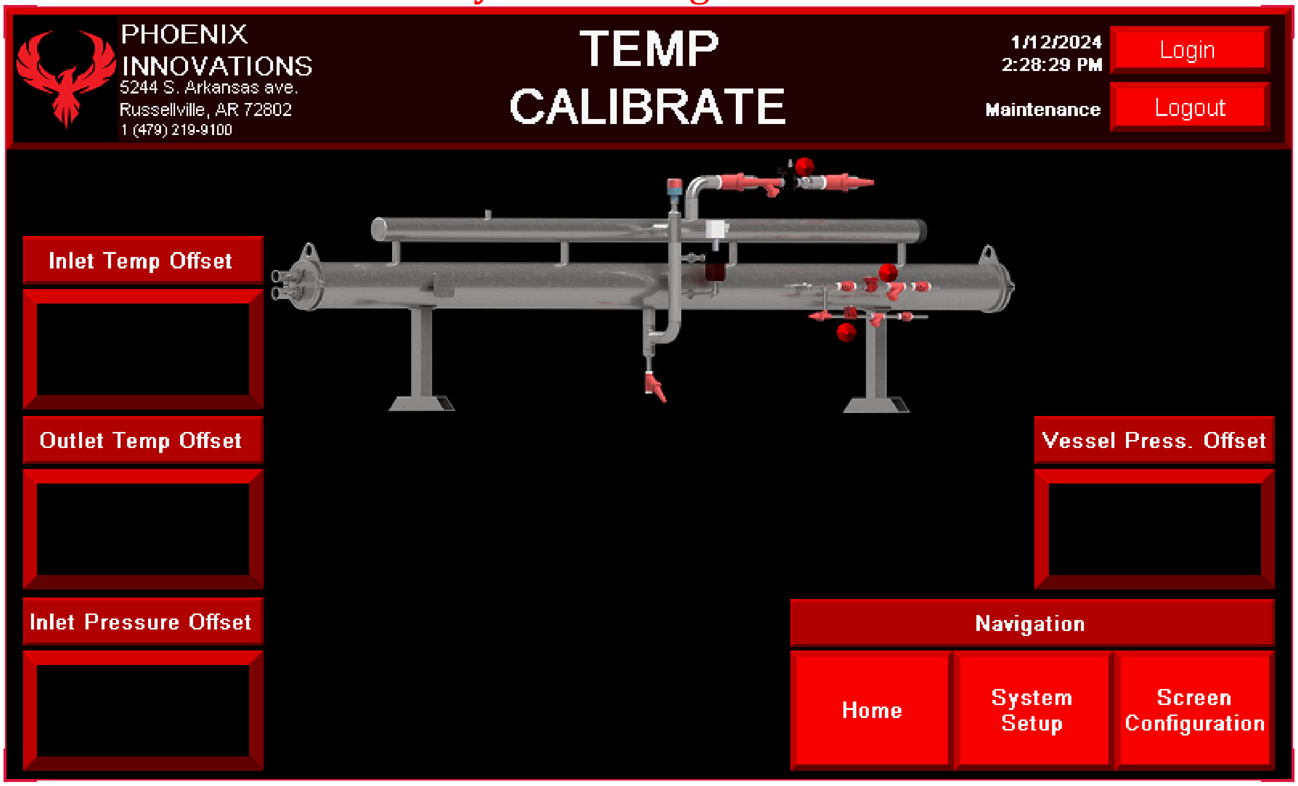

Figure 4 Temp Calibrate Preview

To access the Calibration Screen, you must log in. The Username and Password are both “maint”. The temperatures and pressures can be adjusted here to keep values accurate.

- Inlet Temp Offset – Set the calibration offset of the Inlet temp.
- Outlet Temp Offset – Set the calibration offset of the Outlet temp.
- Inlet Pressure Offset – Set the calibration offset of the Inlet Pressure.
- Navigation – This panel is for navigating to the different screens.
  - Home – Navigates to the home page.
  - System Setup – LOGIN REQ. – Navigates to the first configuration page.
  - Screen Configuration – LOGIN REQ. – Navigates to the built in terminal settings workspace.

System Setup

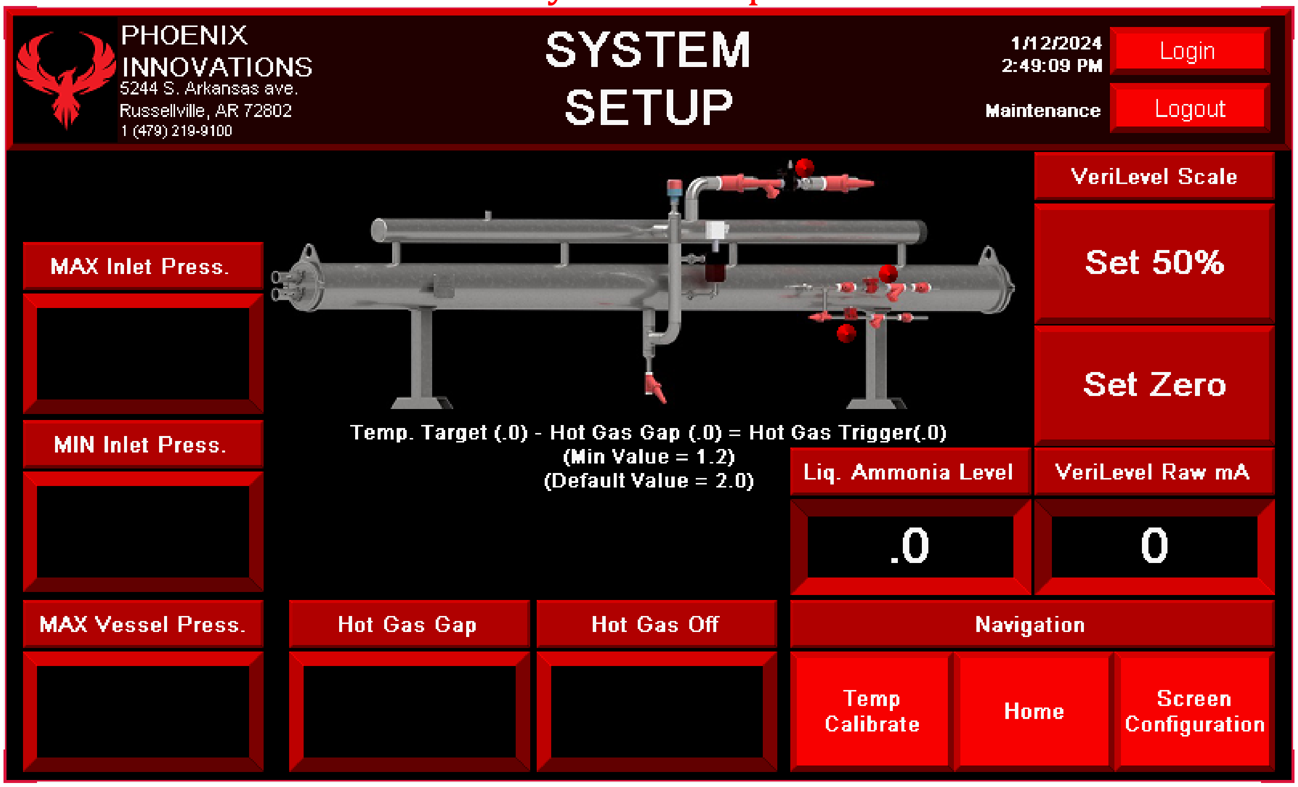

Figure 5 System Setup Preview

Initial setup should typically be done by Phoenix Innovations. That being said, there are situations where access to this screen will be needed, so it is available while logged in with the “maint” login.

- MAX Inlet Press. – Sets the pressure that the system will go into freeze protection.
- MIN Inlet Press. – The minimum inlet pressure keeps the unit from “running dry” while there is no product flowing through the tubes. Due to pump placement, each unit will have a different pressure under normal operation, and this will need to be adjusted accordingly.
- MAX Vesel Press. –
- Hot Gas Gap – The “Hot Gas Gap” is an adjustable offset that determines at what temperature the hot gas is triggered. For example, if the target temperature is set at 34°, and the hot gas gap is set to 2.0, if the product temperature dropped below 32°, the hot gas would cycle to avoid a freeze up.
- Hot Gas Off – The temperature at which the system will come out of Hot Gas Run.
- VeriLevel Scale – Section is used to set the scaling of the VeriLevel sensor using the sight glass on the float column.
  - Set 50% – After zero is set fill the vessel with ammonia until it is in the center of the sight glass.
  - Set Zero – With the vessel completely empty verify that the Raw mA is not 0 and then press the set zero button to set the 0% level.
- Liq. Ammonia Level – The level of ammonia as a percentage of the VeriLevel
- VeriLevel Raw mA – The raw value from the ADC on the analog input for the VeriLevel
- Navigation – This panel is for navigating to the different screens.
  - Temp Calibrate – LOGIN REQ. – Navigates to the temperature / pressure calibration screen
  - Home – Navigates to the home page.
  - Screen Configuration – LOGIN REQ. – Navigates to the built in terminal settings workspace.

Electrical Schematics

Mechanical Layouts

created by

Project Name: 501281 PPV-143
Project Path: C:\Users\jcallahan\Phoenix Innovations\Phoenix Sharepoint - Docu
Plot Date: 3/20/2025
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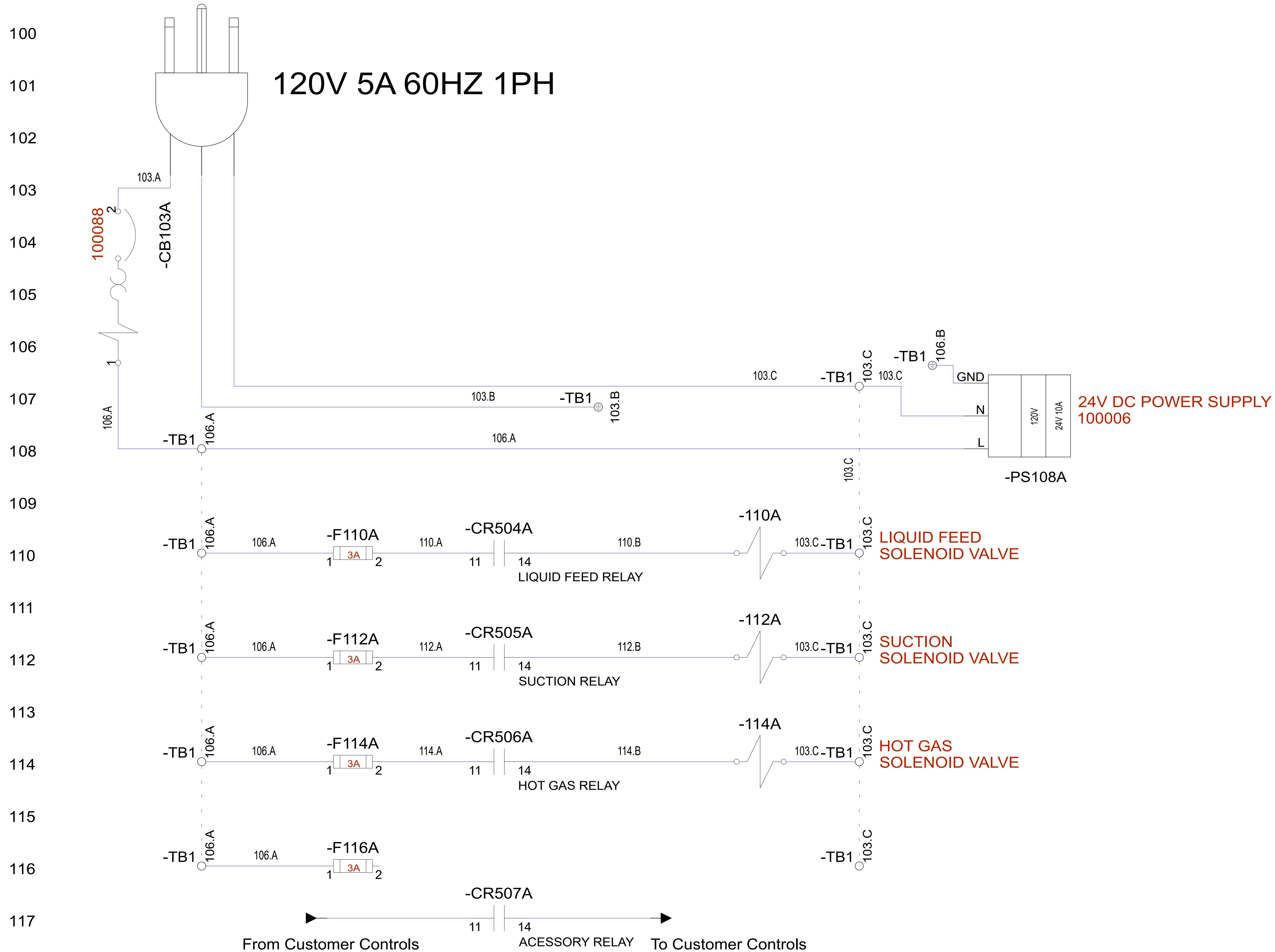
CONTROLS

Status	change	date	name



Sanderson Union Springs, AL SO 1151	PGI-010 **TENDERS HEAT EXCHANGER**	
Repl. by	Origin	Repl. f.
date	SN: PPV-143	
User J. CALLAHAN		
Proved	GSS PARENT # 501281	Sheet 1 of 18 Sh.
Standard		

CONTROLS

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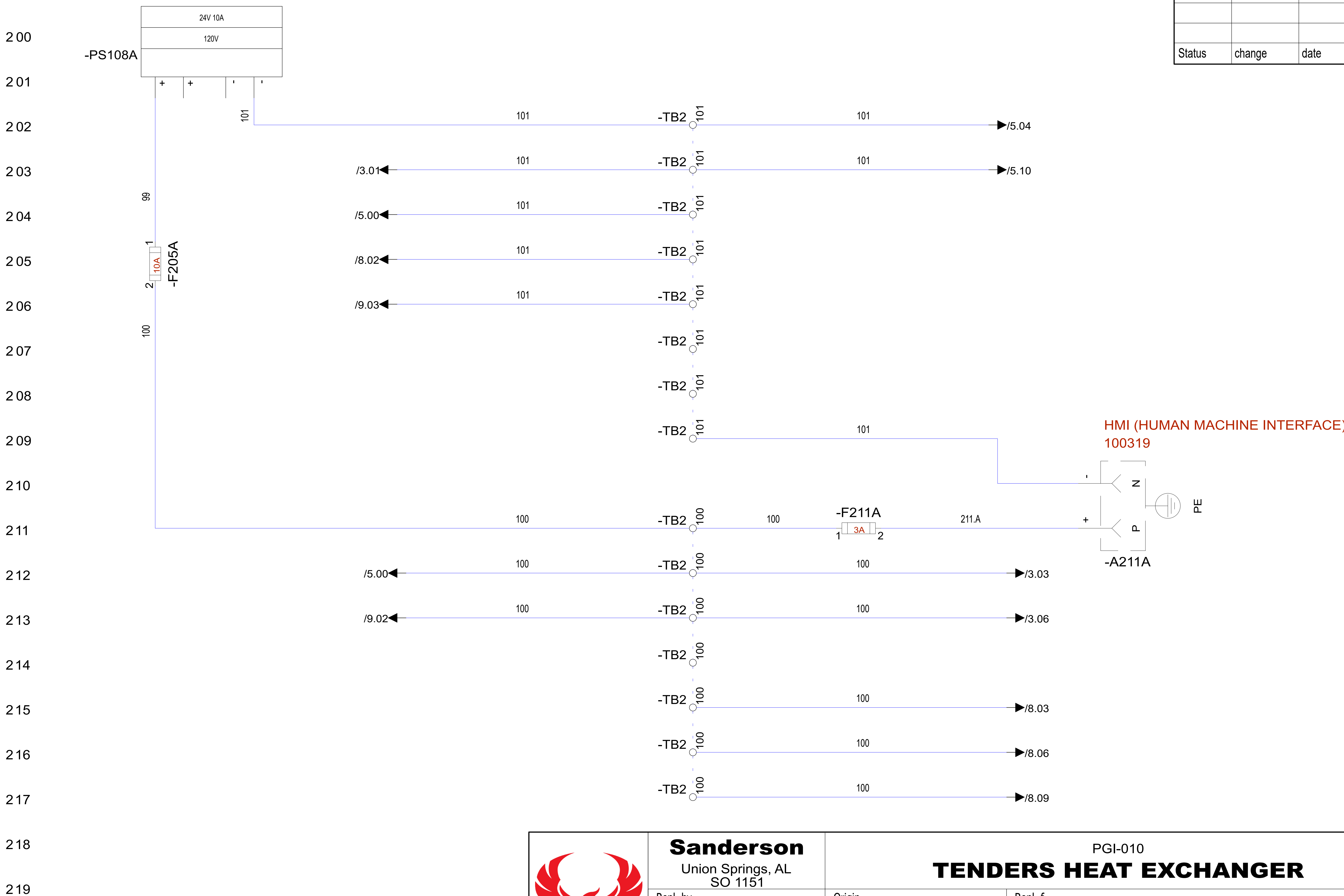
created by

Status	change	date	name



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User J. CALLAHAN		
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Standard		

CONTROLS

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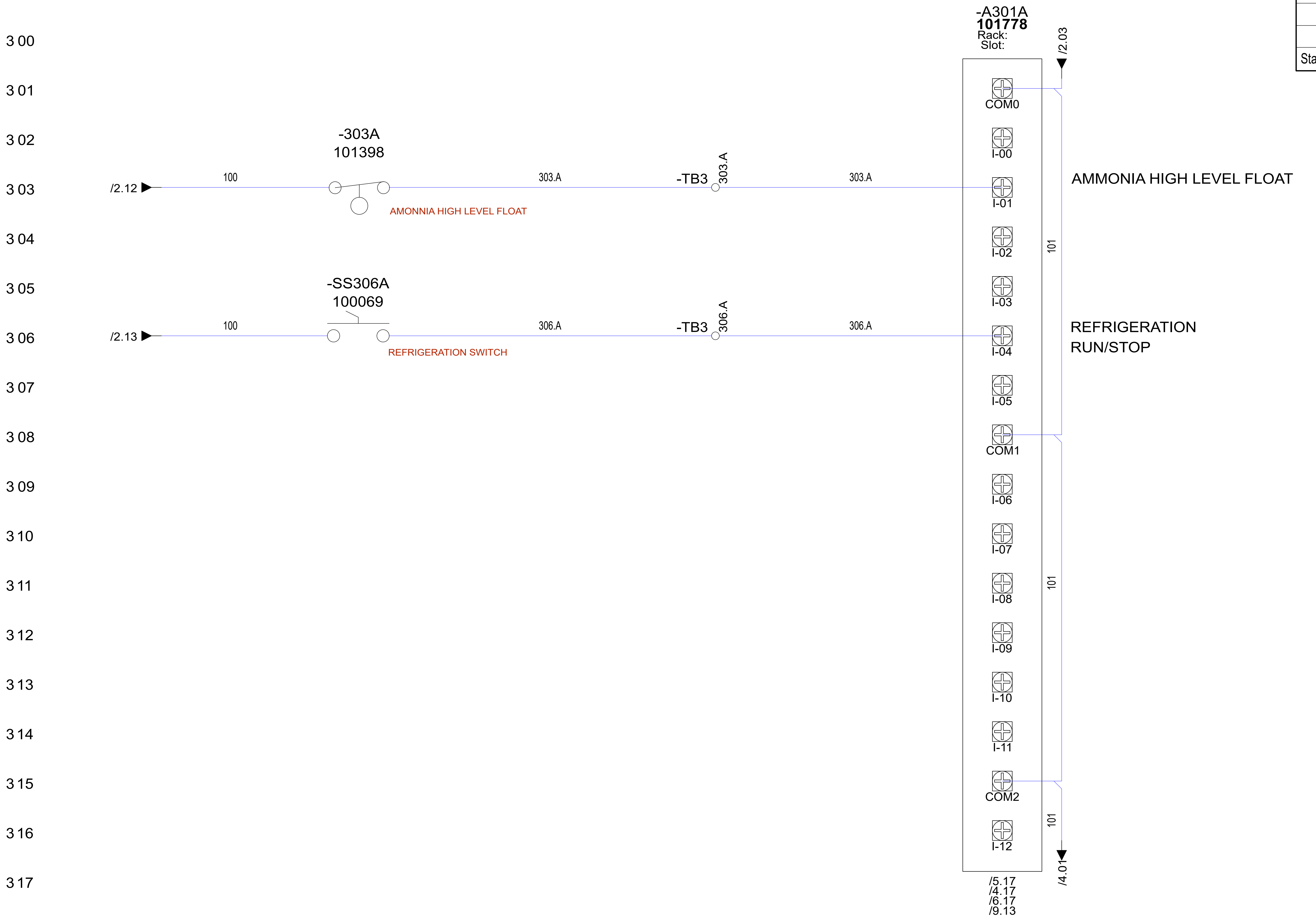
Status	change	date	name



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date	SN: PPV-143	
User J. CALLAHAN		
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CONTROLS

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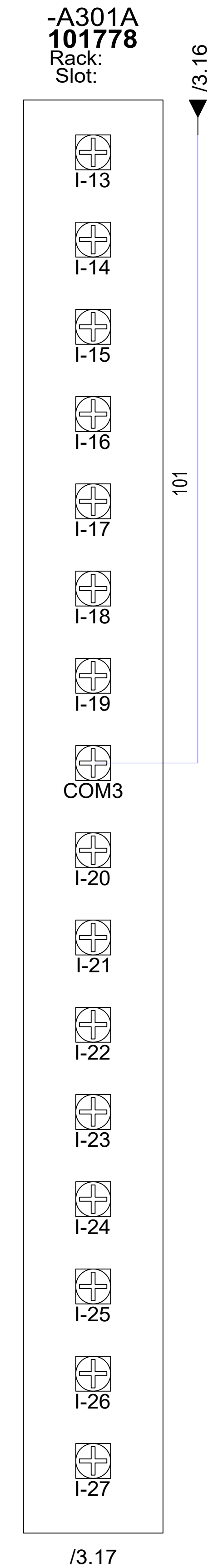

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User	J. CALLAHAN		
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Standard			Sheet 4 of 18 Sh.

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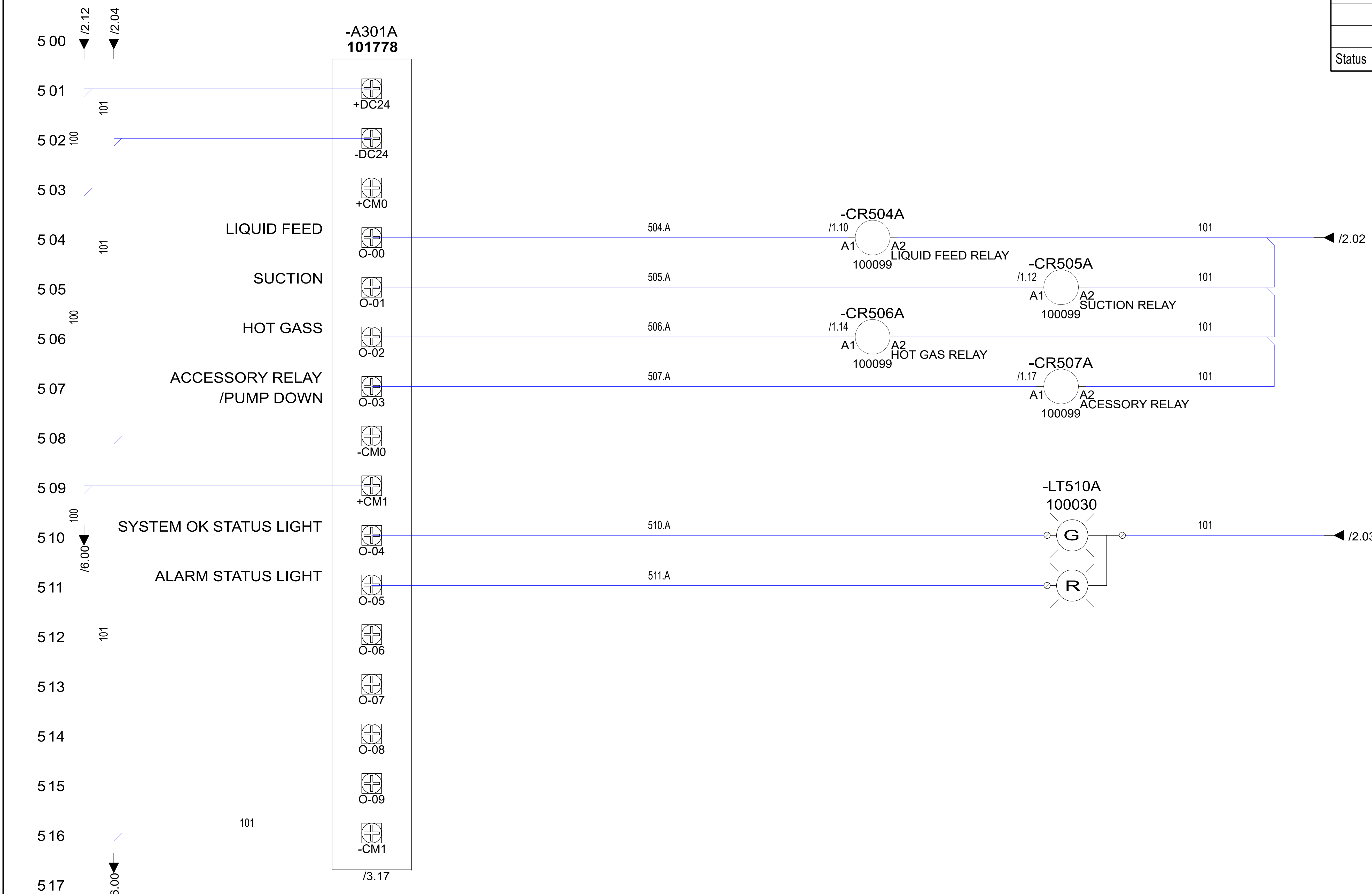

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date			
User	J. CALLAHAN	SN: PPV-143	
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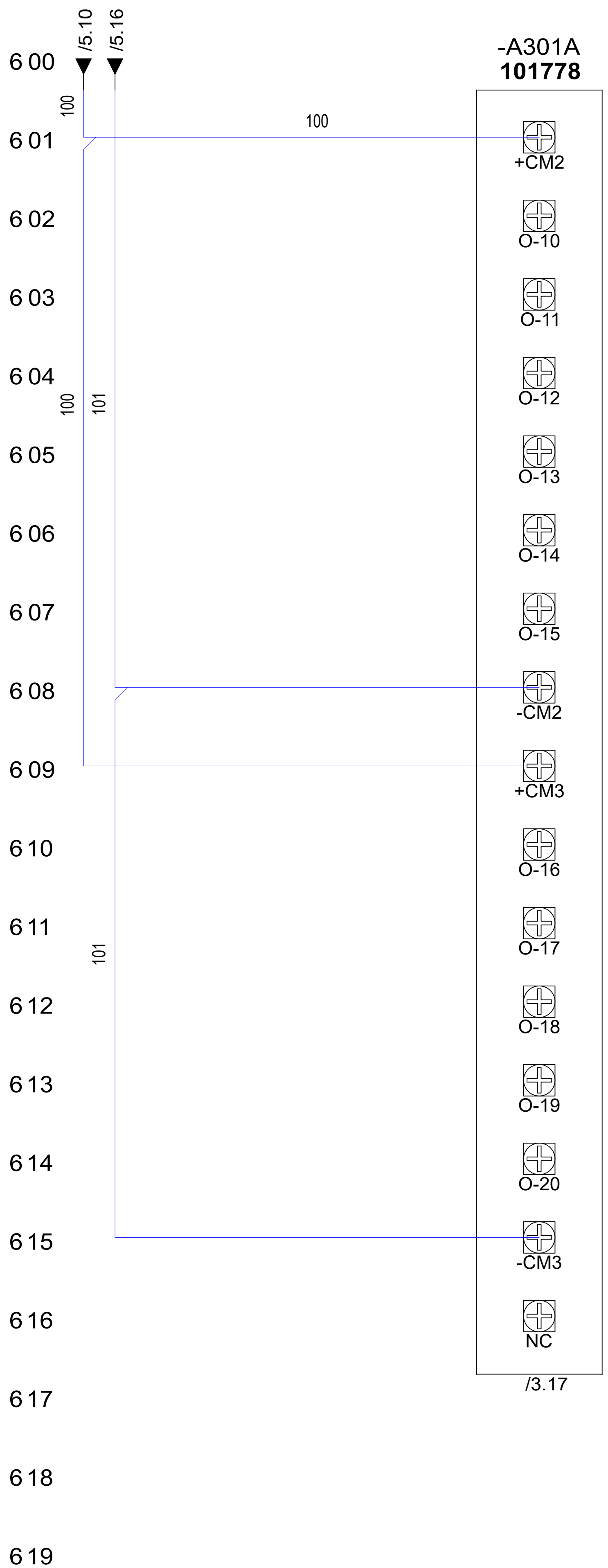

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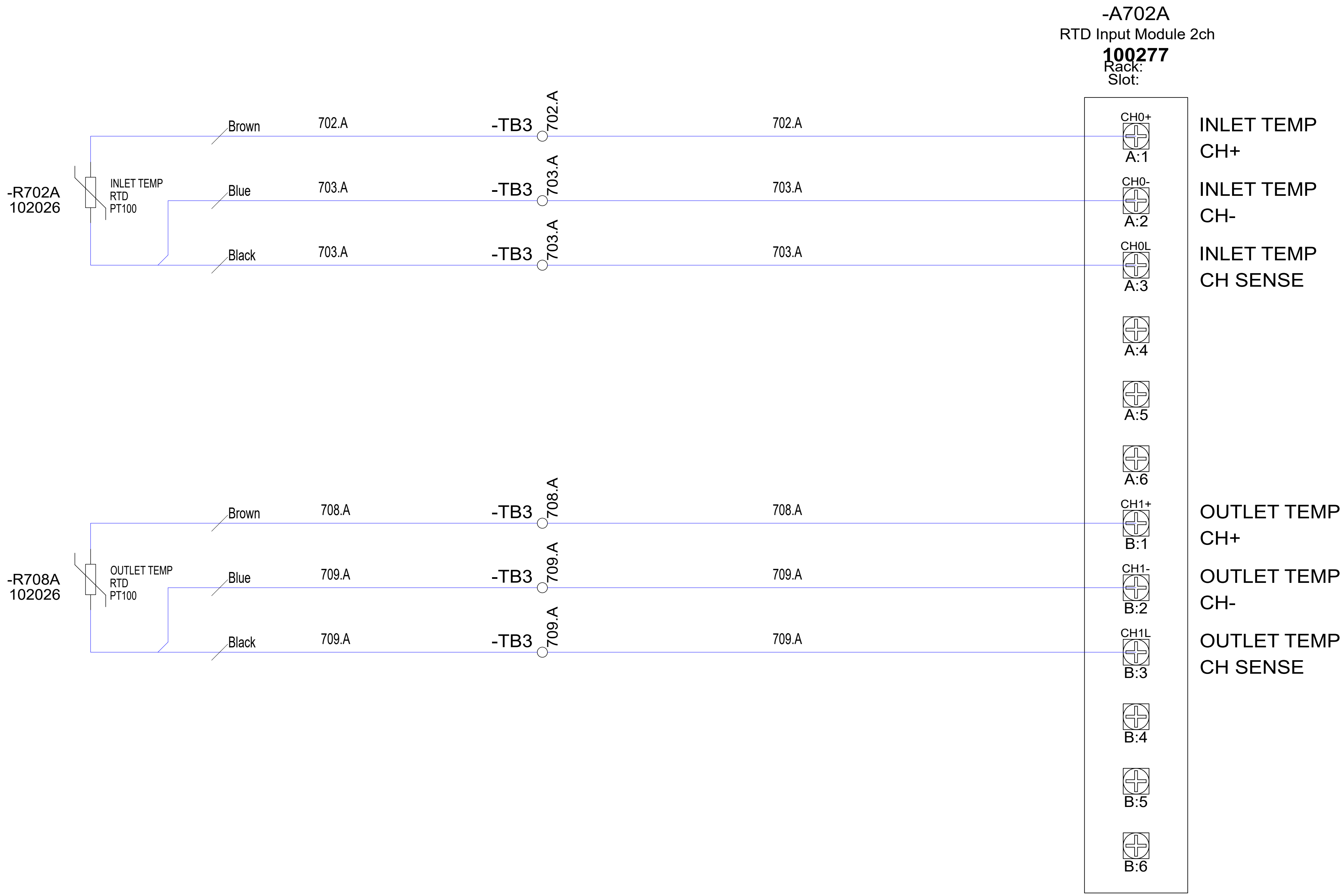

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User	J. CALLAHAN	SN: PPV-143	
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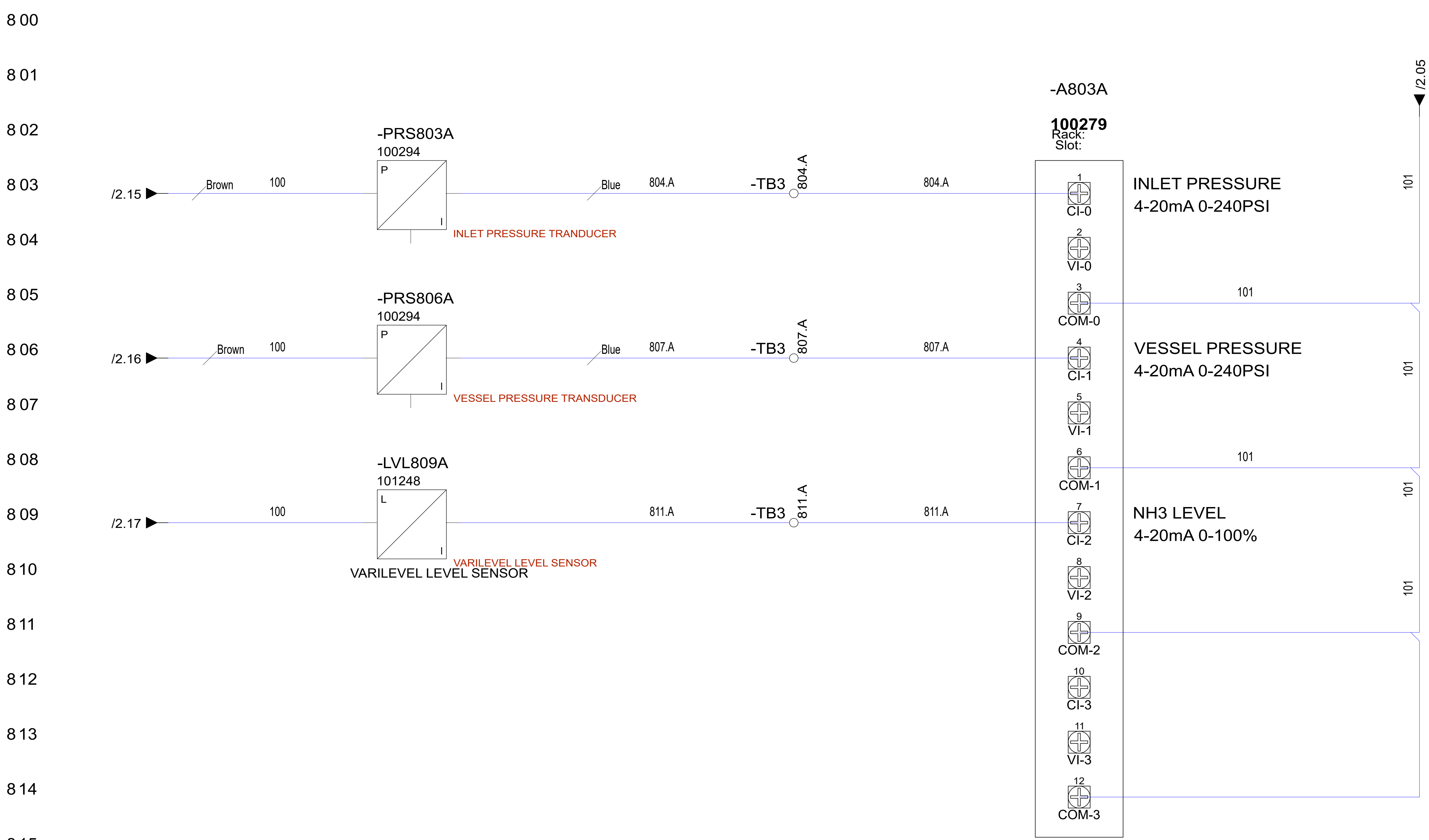

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Repl. by	Origin	Repl. f.	
date			
User J. CALLAHAN	SN: PPV-143		
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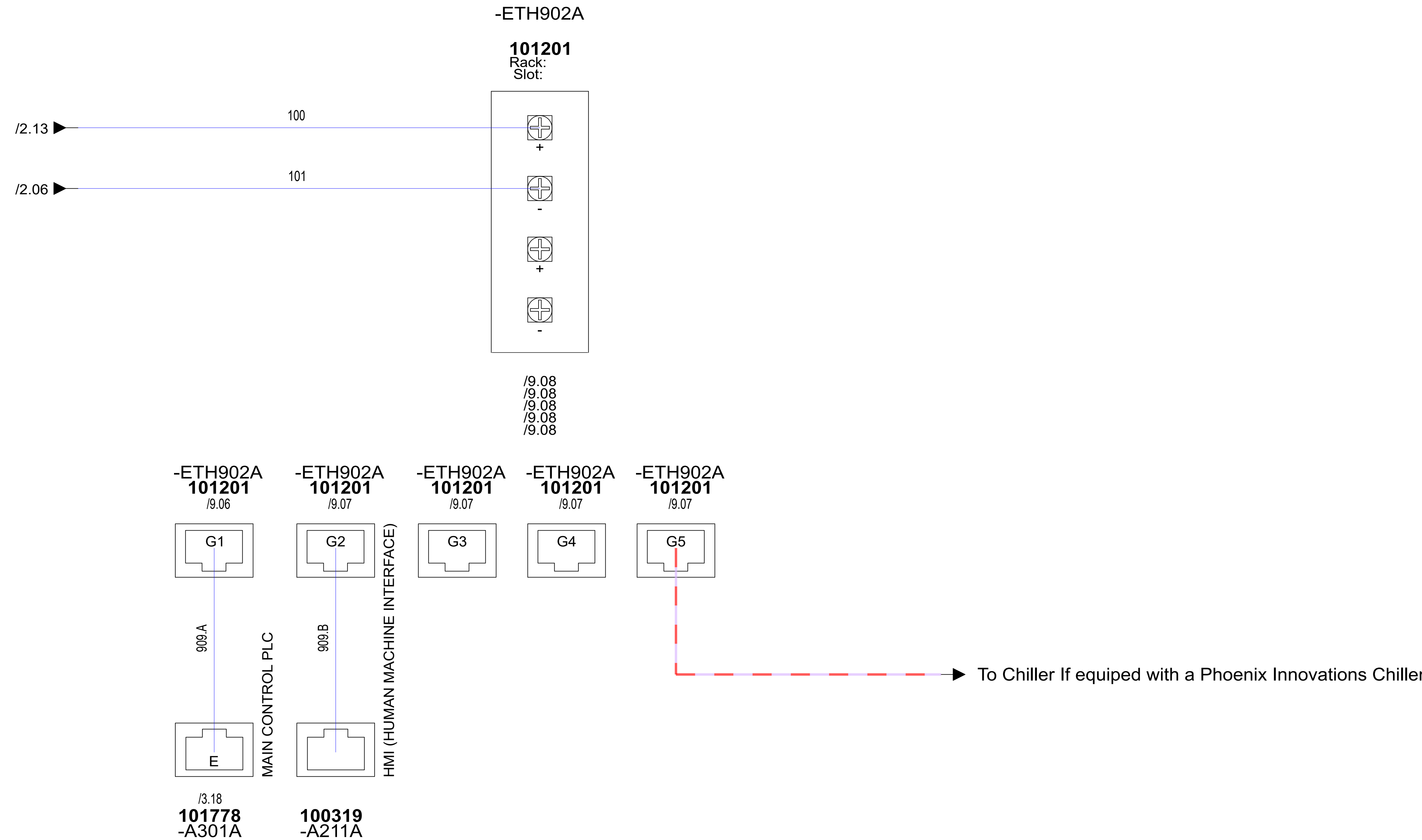

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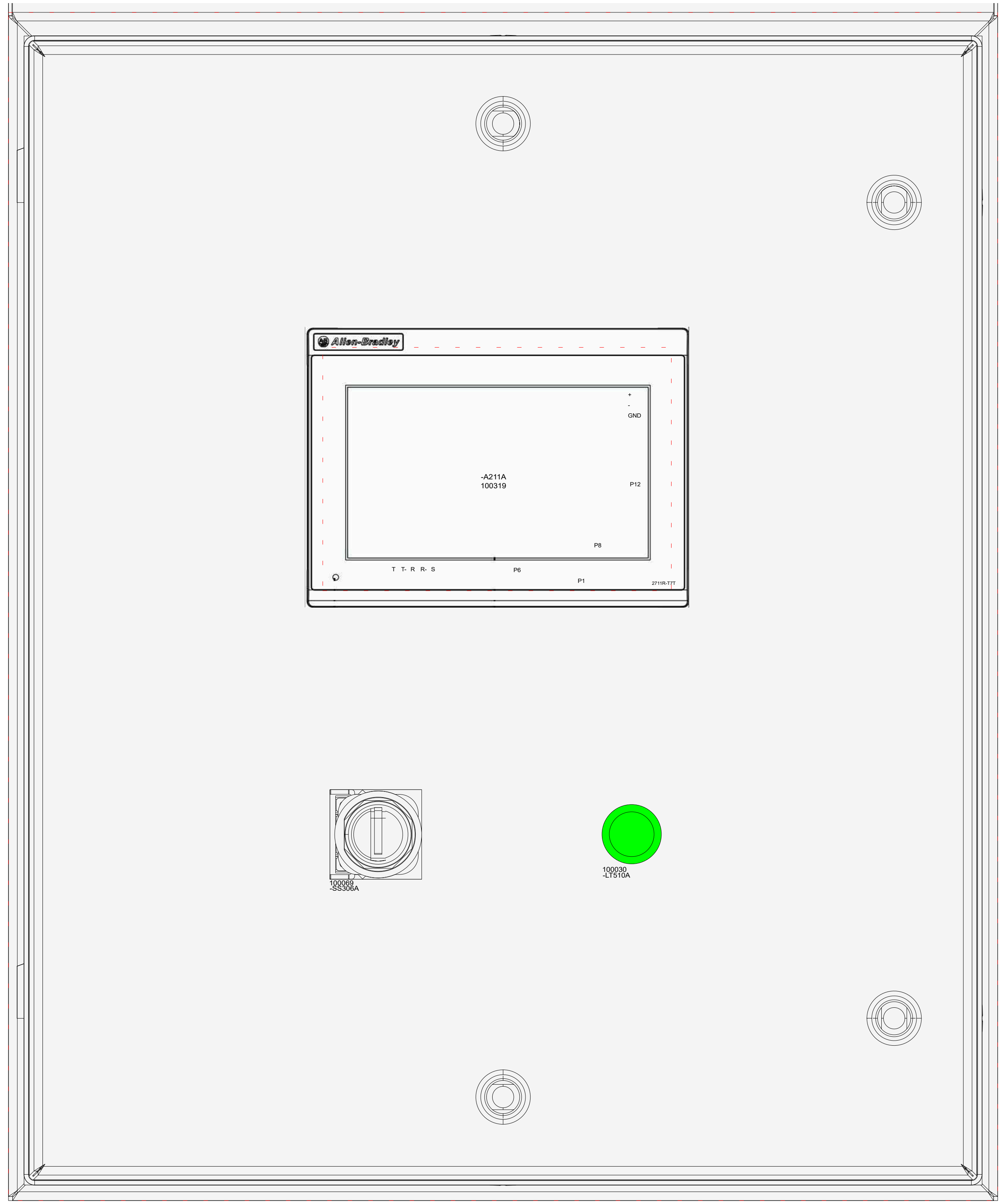

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User: J. CALLAHAN		
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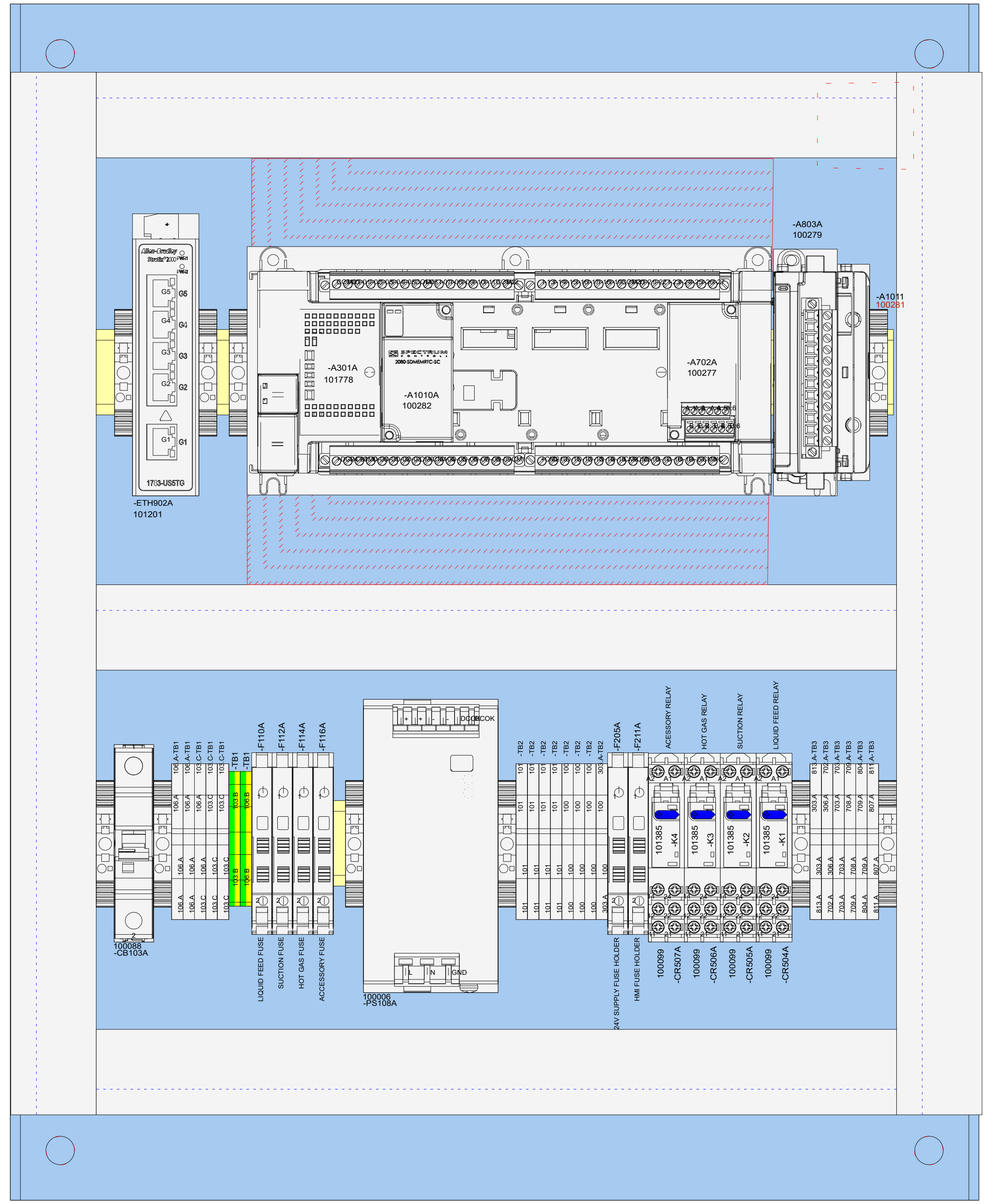

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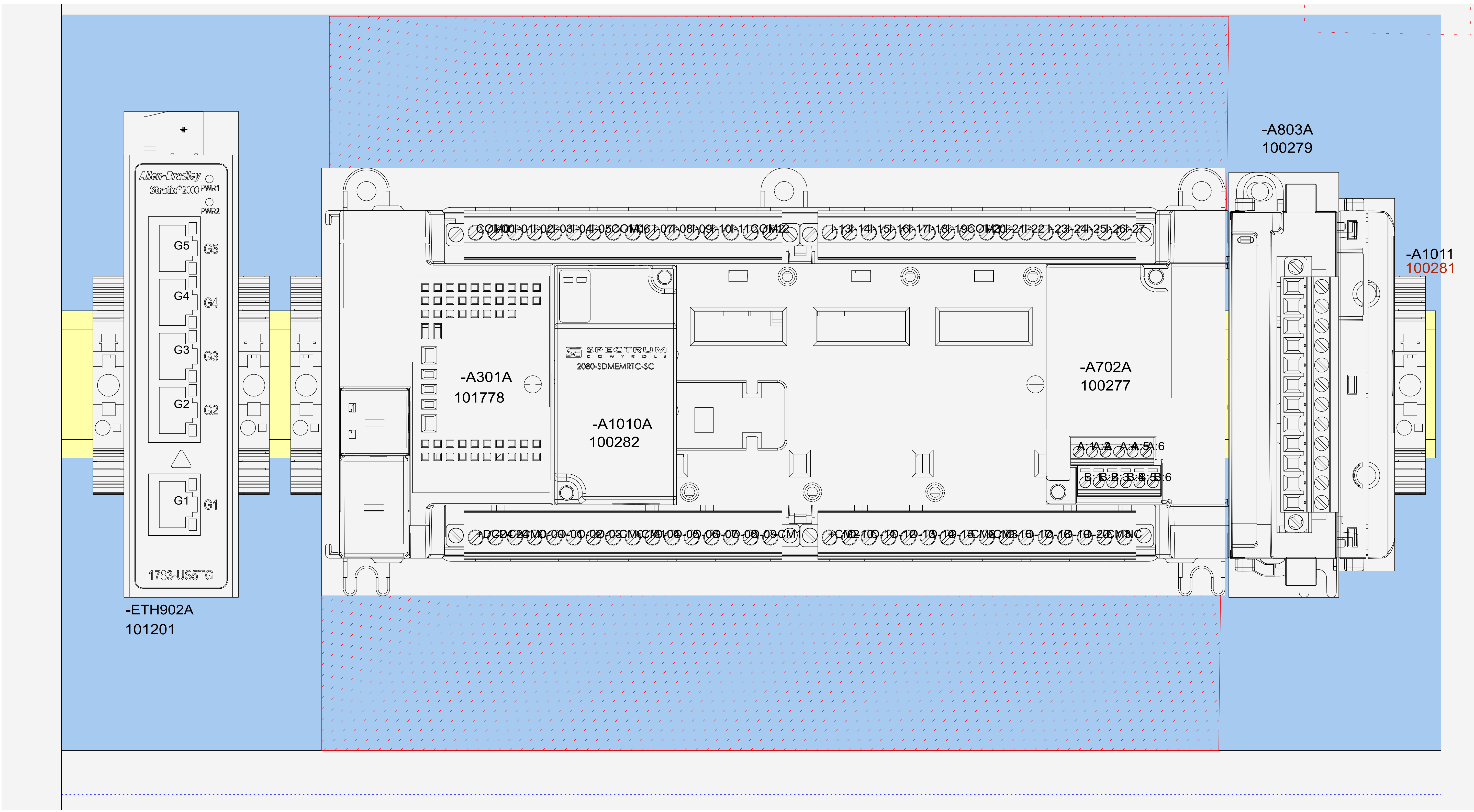

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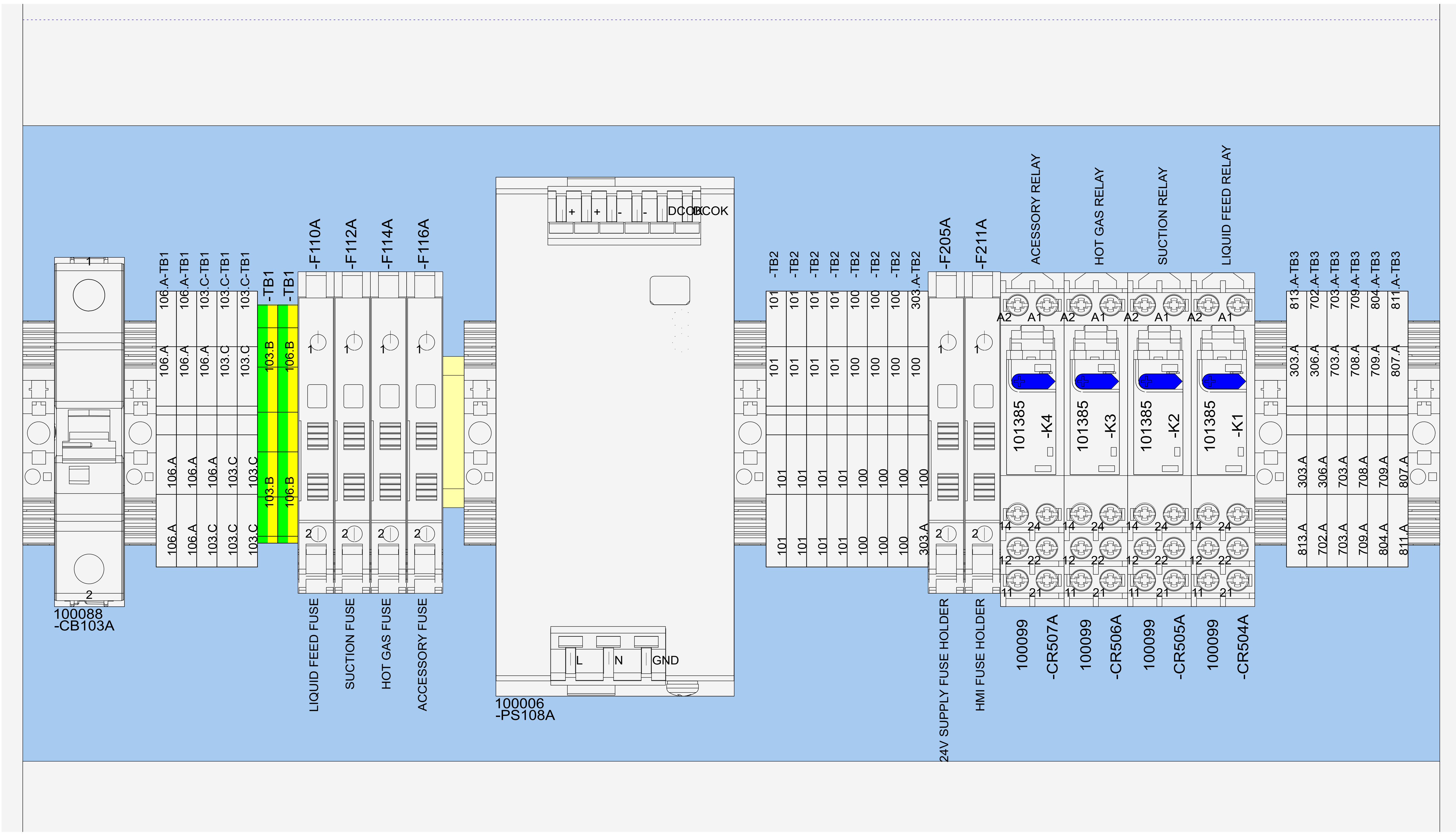

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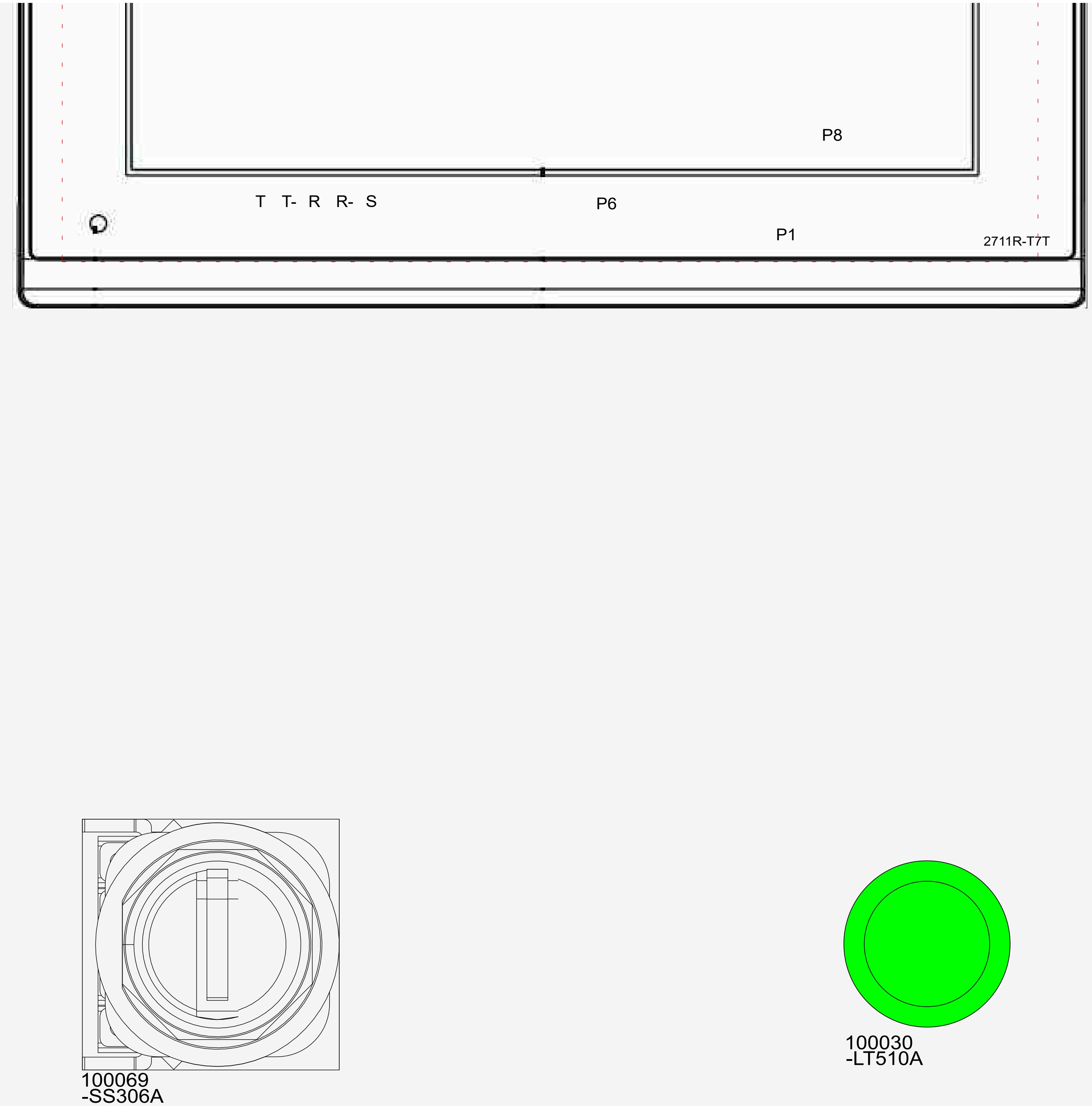

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User: J. CALLAHAN			
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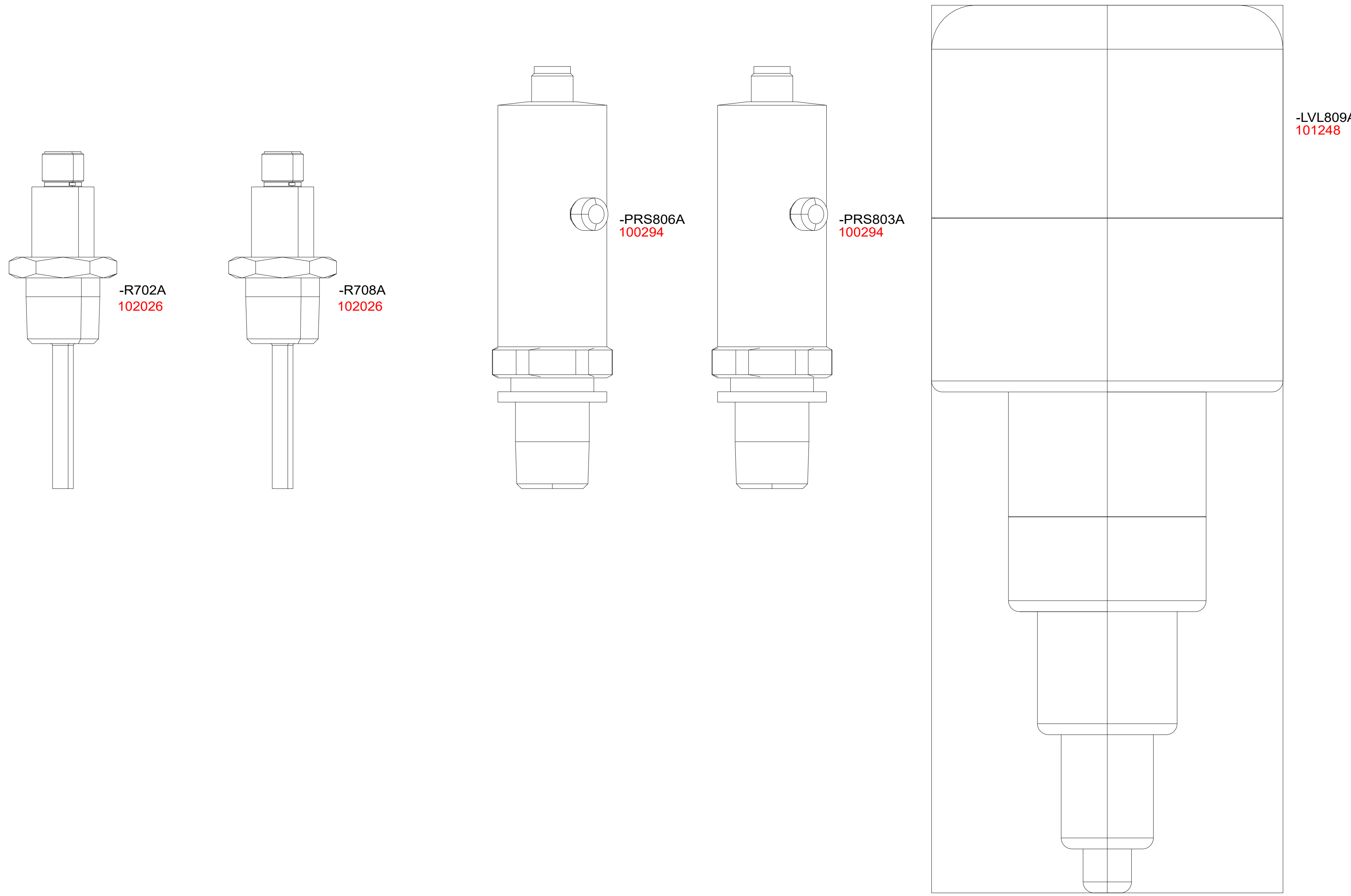

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TENDERS HEAT EXCHANGER

Repl. by		Origin	Repl. f.
date		SN: PPV-143	
User	J. CALLAHAN		
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Standard			

CONTROLS

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Project Path: C:\Users\jcallahan\Phoenix Innovations\Phoenix Sharepoint - Docu
Plot Date: 3/20/2025
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created by

Device/Quantity-Bill of Material with Device Function

Status	change	date	name

Project name: Heat Exchanger With Float Column | Creation date: 28. Feb. 2025 11:21:18 | jcallahan

Amount	PART NUMBER (Component code)	GSS description		Supplier / Manufacturer
	Device designation	Placement	Function	
1	100006	POWERSUPPLY 10A 24V 110VACIN		Rockwell_Automation
	-PS108A	/1.08	24V DC POWER SUPPLY	
1	100030	INDCATOR DUAL COLOR (G&R) 22MM		Rockwell_Automation
	-LT510A	/5.10		
1	100046	ENCLOSURE 24X20X08 HIGH-SHED		nVent HOFFMAN
	-ENC1	/10.17		
1	100069	SWITCH 2-POS 1NO1NC 30MM		Rockwell_Automation
	-SS306A	/3.06	REFRIGERATION SWITCH	
1	100088	CIRCUIT BREAKER 5A 1PH		DTM-E3.series
	-CB103A	/1.03	MAIN FEEDER CIRCUIT BREAKER	
4	100099	8 BLADE RELAY SOCKET		
	-CR504A	/5.04	LIQUID FEED RELAY	
	-CR505A	/5.05	SUCTION RELAY	
	-CR506A	/5.06	HOT GAS RELAY	
	-CR507A	/5.07	ACESSORY RELAY	
1	100277	MICRO800 4CH RTD IN MODULE		Rockwell_Automation
	-A702A	/7.02		
1	100279	MICRO800 4POINTANALOGINPUTEXP		
	-A803A	/8.03		
1	100281	MICRO800-EXPANSIONMODEENDCAP		
	-A1011	/11.07	EXPANSION MODULE ENDCAP	
1	100282	MICRO850 SD CARDADAPTERMODULE		Rockwell_Automation
	-A1010A	/11.07		
2	100294	TRANSDUCER 240PSI 4-20MA NPT		
	-PRS803A	/8.03	INLET PRESSURE TRANDUCER	
	-PRS806A	/8.06	VESSEL PRESSURE TRANSDUCER	
1	100319	TOUCH PANEL PV800 7IN		Allen-Bradley
	-A211A	/2.11	HMI (HUMAN MACHINE INTERFACE)	

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Union Springs, AL
SO 1151

PGI-010
TENDERS HEAT EXCHANGER

Repl. by | Origin | Repl. f.
date
User | SN: PPV-143
Proved | BOM | GSS PARENT # 501281
Standard

Amount	PART NUMBER (Component code)	GSS description		Supplier / Manufacturer
	Device designation	Placement	Function	
4	101193	CABLE 10M M12 5PIN YLW ST		
	-CBL1		INLET RTD CABLE	
	-CBL2		OUTLET RTD CABLE	
	-CBL3		INLET PRESSURE TRANSDUCER CABLE	
	-CBL4		VESSEL PRESSURE TRANSDUCER CABLE	
1	101201	ETHERNET SW 5 PORT		Allen Bradley
	-ETH902A	/9.02		
1	101248	AKS 4100 LIQUID LEVEL SENSOR		
	-LVL809A	/8.09	VARILEVEL LEVEL SENSOR	
4	101385	RELAY DC 24V 8A		
	-K1	/11.11	RELAY CARTRIDGE	
	-K2	/11.11	RELAY CARTRIDGE	
	-K3	/11.11	RELAY CARTRIDGE	
	-K4	/11.11	RELAY CARTRIDGE	
1	101713	COVER LATCHED 10" X 8" HMI		
	-ENC3			
1	101778	Micro800 48 pt controller		Allen-Bradley
	-A301A	/3.01	MAIN CONTROL PLC	
2	102026	1/2" RTD 50 MM INS LNG		
	-R702A	/7.02	INLET TEMP RTD	
	-R708A	/7.08	OUTLET TEMP RTD	
2	501283	BACK PANEL		nVent HOFFMAN
	-ENC2	/11.16		
	-ENC2	/11.16		

Status	change	date	name

CONTROLS

Project Name: 501281 PPV-143
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Repl. by | Origin | Repl. f.
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Standard

SN: PPV-143
BOM
GSS PARENT # 501281

Simplified Complete BOM

jcallahan

Project name:	Heat Exchanger With Float Column	Creation date:	28. Feb. 2025 11:21:20

Amount	PART NUMBER (Component code)	GSS description	Used ✔ / #
1	100006	POWERSUPPLY 10A 24V 110VACIN	
1	100030	INDCATOR DUAL COLOR (G&R) 22MM	
1	100046	ENCLOSURE 24X20X08 HIGH-SHED	
1	100069	SWITCH 2-POS 1NO1NC 30MM	
1	100088	CIRCUIT BREAKER 5A 1PH	
4	100099	8 BLADE RELAY SOCKET	
1	100277	MICRO800 4CH RTD IN MODULE	
1	100279	MICRO800 4POINTANALOGINPUTEXP	
1	100281	MICRO800-EXPANSIONMODEENDCAP	
1	100282	MICRO850 SD CARDADAPTERMODULE	
2	100294	TRANSDUCER 240PSI 4-20MA NPT	
1	100319	TOUCH PANEL PV800 7IN	
4	101193	CABLE 10M M12 5PIN YLW ST	
1	101201	ETHERNET SW 5 PORT	
1	101248	AKS 4100 LIQUID LEVEL SENSOR	
4	101385	RELAY DC 24V 8A	
1	101713	COVER LATCHED 10" X 8" HMI	
1	101778	Micro800 48 pt controller	
2	102026	1/2" RTD 50 MM INS LNG	
2	501283	BACK PANEL	

Status	change	date	name

CONTROLS

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GSS PARENT # 501281